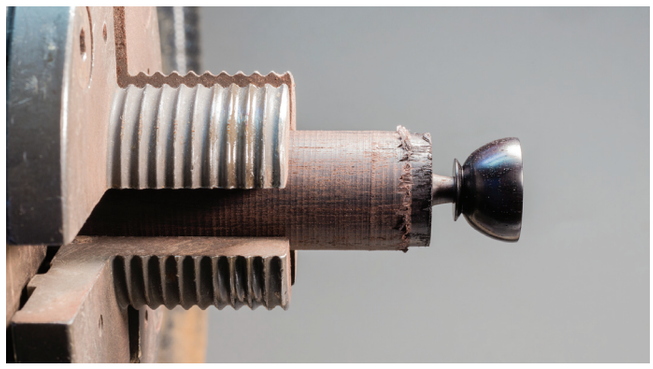
Materials

One of the beauties of turning smaller items, especially miniature sized. You'll enjoy the luxury of not being concerned with material costs. If your end turning is going to be a goblet standing 2mm tall, do you care what the cost per pound or board foot is? Another advantage is that you can use materials that aren't available in bigger pieces. There are many materials that you can find in 25mm³ that regardless of cost, can't be found suitable for turning at 305mm³. When you think about the cost of the materials that wind up on the floor, barring the coring process, turning an exotic wood bowl that is 305mm in diameter produces horrific waste while a 25mm bowl's waste almost doesn't require cleanup. Of course, you can use virtually any species of wood but my recommendation is to branch out from the usual. Wood, even if it is particularly beautiful, doesn't have the same impact in a smaller size; smaller is relative of course. Moving to a 75mm bowl from your typical 610mm bowl really is downsizing. However a 75mm bowl will still show nicely figured wood well. Wood loses its beauty when you are turning a 12mm bowl.

Even the most stunning wood loses its marvel as you get smaller. Aside from wood, you can also choose plastics, bone, antler, precious metals, nuts, and more.

Tagua nuts (*Phytelephas* spp.) work extremely well when miniature turning. The troublesome void when used in a larger turning rarely presents a problem in smaller turnings. If you do decide to use wood, I would recommend using species that are very dense. Small work does turn well from burl that is dense and will often show well depending on your finished size.

My favourites for small turnings are tagua nut, Corian, burls, blackwood (*Dalbergia melanoxylon*), and olivewood (*Olea* spp.). Bone and antler work well too



Blackwood cuts nicely, will hold good detail and can be sanded up to a point where no finish is needed

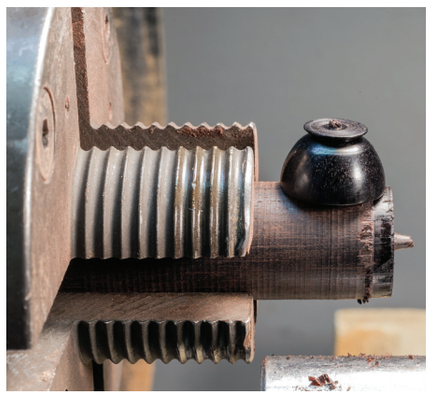
Workholding

As mentioned above, smaller turning workholding lends itself to store bought items at a smaller scale. Virtually all of the full-sized workholding devices, chucks, faceplates and collets, can be purchased in a reduced size. These devices provide the same workholding advantages as the larger versions yet are more appropriately sized with lower mass and the ability to mount on smaller equipment. One of the cautions is to minimize the use of spindle adapters as much as possible. The higher quality adapters run very true but the further your workholding device is located from the headstock spindle face, the more error that can be introduced. This is more noticable with mounting and remounting of the workholding devices but is good practice in general even with high quality adapters. The closer you work to the headstock, the better you'll be from a stability, strength, and accuracy standpoint.

If you do require spindle size adapters, use the best you can find that provide the most true running operation. In addition to the standard mentality of workholding, don't overlook the parent material itself as the workholding mechanism. You can secure the larger piece of your stock in any of the traditional manners and turn that stock to size. Depending on your planned result, you can size stock accordingly to minimise waste. This allows you to use the larger workholding devices for the smaller work on any lathe that the workholding device will fit. This large mass will sometimes be a bit of trouble on deceleration but this can be accommodated. Using the grub screw on chucks and some faceplates will keep the device securely threaded home when turning off the lathe.

Workholding chucks are available for the full-sized lathes through to jewellers lathes. Be cautious of thread adapter runout

Inside collets are sometimes usable but outside collets are preferable. Collets provide for excellent holding and repositioning



Any sized chuck or collet can be used if you are willing to cut away and waste wood. I try to use mounts that hold near finished size

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